SNB

Siam Northeast Bank · Cash Operations

FORECAST-DRIVEN CASH-IN-TRANSIT ROUTING ON DATABRICKS

Zero empty ATMs.
Shorter routes.

Replace a fixed 5-day replenishment calendar with a daily, demand-driven plan for 679 cash points across Khon Kaen.

- **–8%** CIT cost per day
- **–23%** route km
- Cash-out risk **64** → **0** nodes
- **≈ THB 1.5M** a year

THE OUTCOME, IN THE BUYER'S KPIS

Same estate, same day: lower cost, zero cash-outs

CIT COST / DAY

฿69.6K vs ฿75.4K

▼ 8% · ≈ THB 1.5M a year

ROUTE DISTANCE

1,323 km vs 1,711

▼ 23% · 30 fewer crew-hours

CASH-OUT RISK NODES

0 vs 64

Every at-risk machine visited in time

TIME-WINDOW SLA

100% vs 14.7%

Inside bank opening hours

IDLE CASH RELEASED

฿43M

Non-earning float freed

FORECAST ERROR (WAPE)

25% vs 39% naive

Better signal → smaller safety stock

Honest trade-off: the plan dispatches **13 vans instead of 11**. Two extra crews is what it takes to reach SLA and zero cash-outs; total cost still falls because 38 needless stops disappear.

THE PROBLEM

A calendar, not demand, decides where cash goes

SNB replenishes 159 branches, 400 ATMs and 120 cash-recycling machines from 3 cash centres with 30 armoured vans, every 5 days, whatever the demand.

1

Wasted stops

Vans visit machines that are still mostly full because "it's day 5".

FUEL · CREW HOURS · WEAR

2

Cash-outs

Payday and holiday spikes empty ATMs before the next visit.

LOST FEES · COMPLAINTS · TRUST

3

Idle cash

Cassettes and vaults overstocked "just in case".

NON-EARNING FLOAT

4

Static routes

Same sequence every cycle; overruns opening windows.

EXCESS KM · OVERTIME

Cash Ops plans from yesterday's spreadsheets and cannot answer: **"Which machines run dry by Friday, and which van should go?"**

WHO OWNS THE DECISION

Two buyers, one platform

EXECUTIVE SPONSOR · CFO / COO

Cost-to-serve and float

- **Lower CIT cost** per machine: fewer km and stops for the same network
- **Less idle float**: cash that can be redeployed rather than funded
- **Fewer cash-outs**: protected fee income and customer trust
- One governed platform instead of spreadsheets plus point tools

DOMAIN OWNER · HEAD OF CASH OPERATIONS

A plan ready every morning

- Who needs cash, how much, which van, in what order
- A map of every route against the old cycle
- **Override** any stop with a reason, fully audited
- Ask in plain language: "*Which ATMs in Ban Phai run out before Friday?*"
- Re-forecast one machine when an event or holiday is announced

HOW IT WORKS

One integrated data journey: raw feed to dispatcher's screen

01 · INGEST
Lakeflow SDP
Auto Loader on 8 raw feeds in a UC volume · bronze → silver → gold · expectations

›

02 · PREDICT
LightGBM forecast
Point + P80 · MLflow · UC registry `@champion` · Serving `snb-cash-forecast`

›

03 · DECIDE
Orders + OR-Tools
Deliver / pickup, amount, priority, window · CVRP-TW per depot · baseline in same schema

›

04 · SERVE
Lakebase
Synced route stops + node status (ms reads) · writable `dispatch_overrides`

›

05 · ASK
Genie agent
"SNB Cash Ops Assistant" over governed gold tables

›

06 · ACT
Databricks App
`snb-cash-route` · map, KPIs, re-forecast, overrides, Genie chat

Unity Catalog, end to end Layer, domain and PII tags · Custodian name/phone masked · Depot row filter · Gold-only grants · Lineage raw file → app

Each stage consumes the previous stage's governed output. Nothing is copied out of Unity Catalog.

Packaged as one Databricks Asset Bundle: pipeline, job and app defined as code in the repo.

Serverless everywhere: pipeline, jobs, SQL warehouse, model serving (scale to zero), Lakebase Autoscaling.

LAKEFLOW + UNITY CATALOG

Trusted cash data, governed once

LAKEFLOW SPARK DECLARATIVE PIPELINES · SNB_CASH_OPS_PIPELINE

- **Bronze:** Auto Loader ingests 8 feeds (depots, vans, branches, machines, calendar, daily machine and branch cash, CIT trips) incrementally
- **Silver:** typed tables with **expectations**: drop null keys and negative cash; warn on coordinates outside Thailand
- **Gold:** unified node cash flow, demand features, node status and baseline KPIs as materialized views
- Event log and data-quality metrics captured for every update

UNITY CATALOG GOVERNANCE

- **Column mask** on CIT custodian name and phone for anyone outside `snb_pii_readers`
- **Row filter** by depot: a dispatcher in one cash centre sees only that centre's trips
- **Tags** for layer, domain and PII; **gold-only grants** to consumers
- **Lineage** from raw file to the app, visible in Catalog Explorer

Why Databricks: quality rules, security and lineage are declared once and enforced for SQL, ML, Genie and the App alike. No copies, no second security model.

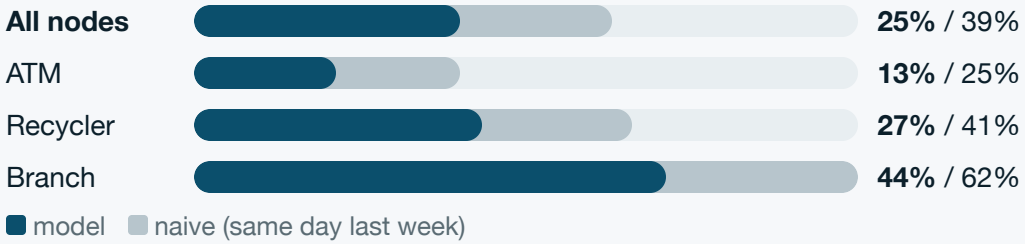
INTELLIGENCE

Forecast demand, then solve the routes

FORECAST · MLFLOW · UC REGISTRY · MODEL SERVING

- **One global LightGBM** across 679 nodes: calendar, payday, holiday, lag and rolling features
- **Point forecast plus a P80 quantile**: replenish for a busy day, not an average day
- `cash_demand_forecast@champion` served on `snb-cash-forecast`; the App's re-forecast calls the same endpoint

HOLDOUT WAPE (AUG–SEP 2026) · LOWER IS BETTER



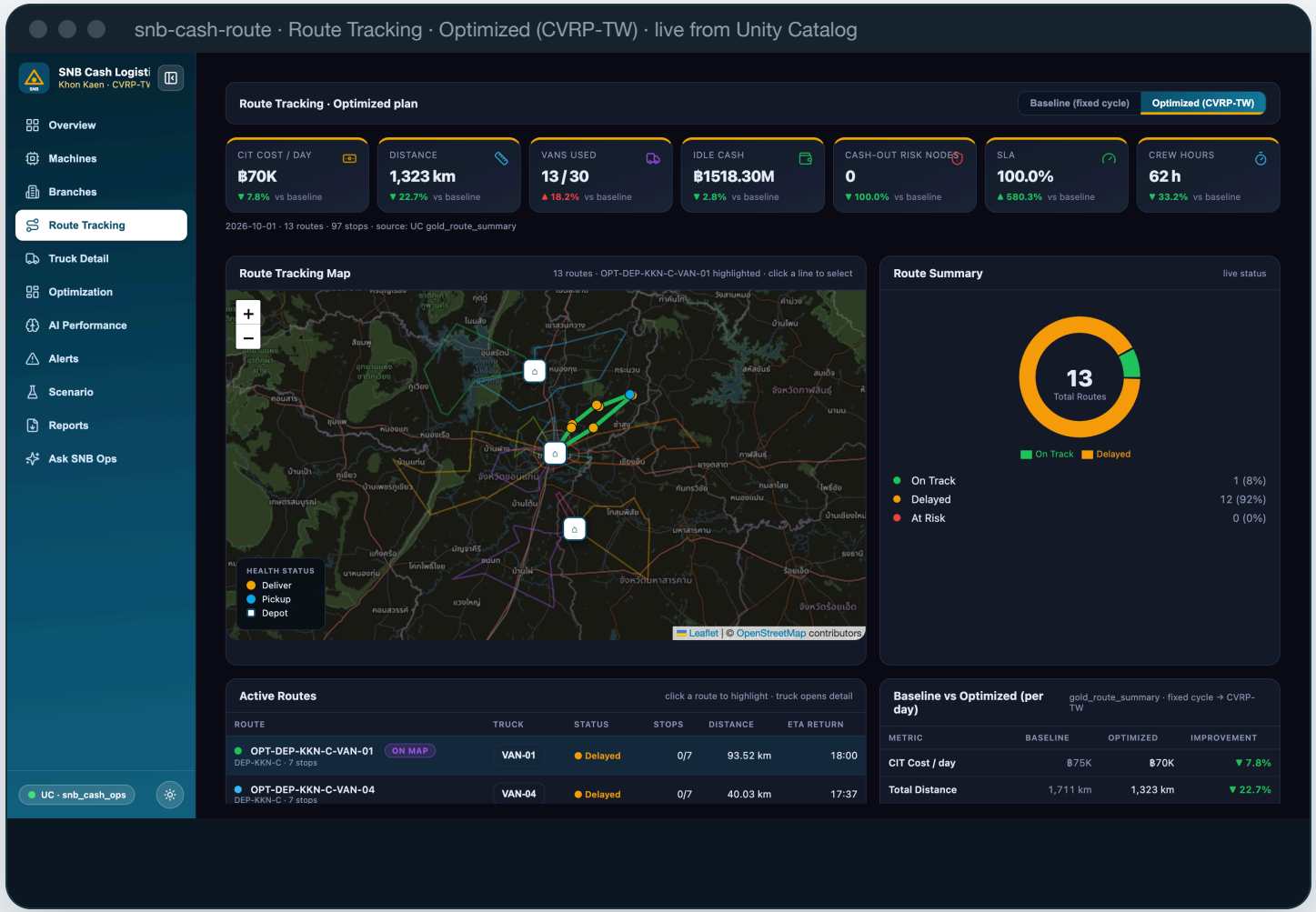
OPTIMIZE · OR-TOOLS CVRP-TW

- Forecast → **97 replenishment orders**: 75 critical deliveries, 4 normal deliveries, 18 pickups of surplus cash
- Capacitated vehicle routing with time windows **per depot**: van cash capacity, branch opening hours, 07:00–18:00 shift
- 25-second solve per depot; critical stops carry a very high drop penalty, so they are effectively never skipped
- Writes the **fixed-cycle baseline in the same schema**, so every KPI is an apples-to-apples comparison

Why Databricks: features, model, registry, serving and optimizer share the same governed tables.

IN THE DISPATCHER'S HANDS

The control room: Databricks App snb-cash-route



1. **KPI cards vs baseline**
   Cost, km, vans, idle cash, cash-out risk, SLA, crew hours from `gold_route_summary`.

2. **Route map**
   Toggle fixed cycle vs optimized; click a route to see its stops.

3. **Re-forecast**
   Calls the serving endpoint for a 7-day P80 on any machine.

4. **Override with a reason**
   Skip, add, resequence or emergency; written to Lakebase.

5. **Ask SNB Ops**
   Embedded Genie chat over the same governed tables.

LAKEBASE + GENIE

Operational reads, audited writes, plain-language answers

LAKEBASE AUTOSCALING · PROJECT SNB-CASH-OPS

- **UC synced tables** `lb_node_status` (679 rows) and `lb_route_stops` (232 rows) give the App millisecond reads
- Writable Postgres table `dispatch_overrides`: SKIP / ADD / RESEQUENCE / EMERGENCY, with reason and author
- App service principal has least-privilege access to warehouse, endpoint, Lakebase and Genie

A dispatcher's override lands in Postgres with a reason, **not in a chat screenshot**.

GENIE · SNB CASH OPS ASSISTANT (AGENT MODE)

Which nodes run out of cash in the next 3 days if we do nothing?

On **2026-10-01**, **79 nodes** are flagged to run out of cash within 3 days, all marked **CRITICAL** for a **DELIVER** action.

Generated SQL over `gold_replenishment_orders` · masks and row filters still apply

Six benchmark questions with generated SQL and answers are committed in `evidence/genie_benchmarks.txt`.

DEMO FLOW · 7–10 MINUTES

Outcome first, then under the hood

1. **Outcome**
   App KPI cards: cost per day, km, vans, cash-out risk, SLA.

2. **Map**
   Toggle BASELINE vs OPTIMIZED for 1 Oct 2026.

3. **Re-forecast**
   A payday ATM: the serving endpoint returns a 7-day P80.

4. **Override**
   Skip a healthy stop with a reason; the row lands in Lakebase.

5. **Genie**
   "Which ATMs are at cash-out risk tomorrow, and which van covers them?"

6. **Under the hood**
   Pipeline graph and expectations → UC lineage, mask, row filter → MLflow run and endpoint.

7. **Close**
   THB 1.5M a year and the pilot ask.

DECISIONS AND TRADE-OFFS

What we chose, and what it costs

DECISION	WHY	TRADE-OFF
Declarative pipeline, not notebooks	Expectations, lineage, incremental by default	Less ad-hoc flexibility
One global LightGBM, not 679 models	Shares signal, handles new ATMs, one endpoint	Less per-node tuning
P80 quantile for replenishment	A cash-out costs more than a little extra cash	Some extra float carried
OR-Tools, 25 s per depot	Open source, explainable, fast	Not provably optimal
Lakebase synced tables for the App	Fast reads plus writable overrides	Snapshot freshness, not streaming
Genie agent mode, not a custom agent	Governed by UC, no prompt plumbing, fast to value	Less custom tool control
Serverless everywhere	No cluster ops, scale to zero for a once-a-day plan	Cold starts

Full rationale in `docs/decisions.md`.

BUSINESS CASE · SYNTHETIC-DATA DEMO, INDICATIVE

THB 1.5M a year in CIT cost, before the float and cash-out upside

LEVER	BASIS	VALUE
CIT operating cost	(baseline – optimized) cost per day × 365	**≈ THB 1.5M a year**
Idle cash	Float reduction across 679 nodes on the plan day	**THB 43M released**, plus carry cost avoided
Cash-out risk	Risk nodes per plan day, 64 → 0	**Fee income and complaints protected**
Fleet	Vans 11 → 13 per day	Two extra crews take SLA from **15%** → **100%**

Cost model assumptions

Distance cost plus THB 2,500 per van-day; road km = straight-line × 1.3. Synthetic assumptions, re-validated on real SNB data in the pilot before any commitment.

Time to value

The stack deploys from one bundle; a one-cash-centre pilot needs about 2 weeks of feed setup, then 30 CIT days of measurement.

PATH TO PRODUCTION

Pilot one cash centre, then scale and harden

PHASE 1 · 30 CIT DAYS

Pilot

Khon Kaen Central cash centre; real ATM switch, vault and CIT feeds under the same silver contract.

Exit: forecast beats current planner; ≥ 80% of planned stops accepted

PHASE 2

Scale

All 3 cash centres; nightly plan plus a mid-morning re-forecast on payday.

Exit: cost per day and cash-out KPIs tracked weekly

PHASE 3

Harden

IdP groups for row filters, maker-checker on overrides, CI/CD for the bundle, model monitoring, road-network travel times.

Exit: Risk and Security sign-off; PDPA review

PHASE 4

Extend

Denomination mix, vault planning, CIT vendor SLAs, other regions.

Exit: region-by-region rollout

Humans confirm every plan; nothing moves cash autonomously. Core banking, CIT vendors and radio dispatch stay in place.

THE ASK

Pilot one cash centre for 30 CIT days

Khon Kaen Central, measured against the fixed cycle on the same nodes.

≥ 20%
lower CIT cost or km

No more
cash-out incidents than today

≥ 80%
of recommended stops accepted by dispatchers

Next step: a 2-hour working session with Cash Ops and IT Security to scope the pilot data feeds.

Repo: https://github.com/nan-databricks/fe-bar-cash-route